JEEVAN RAKSHA PHARMACY

REST API Design Specification

Base URL: http://localhost:8080

MySQL Database | 38 Endpoints | 7 Domains

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API Architecture Overview

The Jeevan Raksha Pharmacy REST API is a business-complete, resource-oriented HTTP API exposing all operations on the Jeevan Raksha Pharmacy database. The base URL for all endpoints is:

```
http://localhost:8080
```

Domain Summary

Method	Endpoint URL	Purpose
GET/POST/PUT/DELETE	/api/customers/**	Customer registration, profiles, order history
GET/POST/PUT/DELETE	/api/suppliers/**	Supplier management and profiles
GET/POST/PUT/DELETE	/api/medicines/**	Medicine catalog — CRUD, filters, expiry
GET/POST/DELETE	/api/orders/**	Order placement, history, invoices
GET/PATCH	/api/inventory/**	Stock tracking and restocking alerts

Method	Endpoint URL	Purpose
GET	/api/reports/**	Revenue, bestsellers, analytics
GET	/api/search	Cross-entity full-text search

Standard Response Envelope

Every endpoint wraps its result in a consistent JSON envelope so all clients can handle success and failure uniformly.

Success envelope

```
{
  "status": "success",
  "data":   { ... }
}
```

Error envelope

```
{
  "status": "error",
  "code":    "MEDICINE_NOT_FOUND",
  "message": "No medicine found with ID: 99",
  "timestamp": "2024-06-10T14:30:00",
  "errors": [
    { "field": "medicineId", "issue": "must be a valid integer" }
  ]
}
```

Common Error Codes

Error Code	Meaning
CUSTOMER_NOT_FOUND	No customer exists for the given ID

Error Code	Meaning
SUPPLIER_NOT_FOUND	No supplier exists for the given ID
MEDICINE_NOT_FOUND	No medicine exists for the given ID
ORDER_NOT_FOUND	No order exists for the given ID
INSUFFICIENT_STOCK	Ordered quantity exceeds available medicine stock
SUPPLIER_HAS_MEDICINES	Cannot delete supplier — medicines are linked
MEDICINE_IN_ORDER	Cannot delete medicine — it is part of existing orders
DUPLICATE_PHONE	Customer phone number already registered
VALIDATION_ERROR	Request body failed validation (HTTP 400)
INTERNAL_SERVER_ERROR	Unhandled server-side error (HTTP 500)

Pagination Conventions

All list endpoints accept these standard query parameters:

- page — zero-based page index (default: 0)
- size — records per page (default: 10)
- sort — field to sort by (e.g. name, price)
- order — sort direction: asc or desc (default: asc)

Paginated responses include: totalElements, totalPages, page, size alongside the data array.

Domain 1 — Customers

Base path: http://localhost:8080/api/customers

The Customers domain manages patient and buyer registration. Full CRUD with referential integrity on deletion — if orders are linked, the delete is rejected. The phone field acts as the natural deduplication key and must be unique across all customers.

GET	**http://localhost:8080/api/customers**	*List all customers — filterable by city and paginated*

Query params: page, size, sort, order, city

Sample: GET /api/customers?city=Mumbai&page=0&size=10

```
{
  "status": "success",
  "totalElements": 2,
  "page": 0,
  "size": 10,
  "data": [
    { "customerId": 1, "name": "Rahul Sharma",  "phone": "9876543210", "city": "Mumbai" },
    { "customerId": 5, "name": "Vikram Singh",   "phone": "7766554433", "city": "Mumbai" }
  ]
}
```

GET	http://localhost:8080/api/customers/{customerId}	*Customer profile with order summary*

200 OK *Response Body*

```
{
  "status": "success",
  "data": {
    "customerId":       1,
    "name":             "Rahul Sharma",
    "phone":            "9876543210",
    "city":             "Mumbai",
    "totalOrders":      2,
    "totalAmountSpent": 950.00
  }
}
```

404 Not Found *Response Body*

```
{
  "status":  "error",
  "code":    "CUSTOMER_NOT_FOUND",
  "message": "No customer found with ID: 99"
}
```

POST	http://localhost:8080/api/customers	Register a new customer account

Request Body

```
{
  "name":  "Kiran Desai",
  "phone": "9011223344",
  "city":  "Pune"
}
```

201 Created	Response Body

```
{
  "status":  "success",
  "message": "Customer registered successfully",
  "data": {
    "customerId": 6,
    "name":       "Kiran Desai",
    "phone":      "9011223344",
    "city":       "Pune"
  }
}
```

PUT	http://localhost:8080/api/customers/{customerId}	Update customer profile — city or phone change

Request Body

```
{
  "name":  "Kiran Ramesh Desai",
  "phone": "9011223344",
  "city":  "Nashik"
}
```

200 OK	Response Body

```
{
  "status":  "success",
  "message": "Customer profile updated",
  "data":    { "customerId": 6, "city": "Nashik" }
}
```

DELETE	**http://localhost:8080/api/customers/{customerId}**	*Remove customer — blocked if orders are linked*

200 OK *Response Body*

```
{ "status": "success", "message": "Customer with ID 6 deleted successfully" }
```

409 Conflict *Response Body*

```
{
  "status":  "error",
  "code":    "CUSTOMER_HAS_ORDERS",
  "message": "Cannot delete customer with existing orders. Cancel orders first."
}
```

GET	**http://localhost:8080/api/customers/{customerId}/orders**	*Full order history — My Orders page*

```
{
  "status":       "success",
  "customerId":    1,
  "customerName": "Rahul Sharma",
  "totalOrders":   2,
  "data": [
    {
      "orderId":     1,
      "orderDate":   "2023-10-01",
      "orderTotal":  150.00,
      "paymentMode": "UPI",
      "items": [
        { "medicineId": 1, "medicineName": "Dolo 650", "quantity": 5, "subtotal":
150.00 }
      ]
    },
    {
      "orderId":     4,
      "orderDate":   "2023-10-10",
      "orderTotal":  800.00,
      "paymentMode": "UPI",
      "items": [
        { "medicineId": 5, "medicineName": "Insulin Pen", "quantity": 1, "subtotal":
800.00 }
      ]
    }
  ]
}
```


GET	**http://localhost:8080/api/customers/top-spenders**	*Top customers by total amount spent — loyalty programs*

Query param: limit (default 5)

```
{
  "status": "success",
  "data": [
    { "customerId": 4, "name": "Sneha Reddy",   "city": "Hyderabad",
"totalAmountSpent": 1190.00, "totalOrders": 1 },
    { "customerId": 1, "name": "Rahul Sharma",  "city": "Mumbai",
"totalAmountSpent":  950.00, "totalOrders": 2 }
  ]
}
```

Domain 2 — Suppliers

`Base path:  http://localhost:8080/api/suppliers`

The Suppliers domain manages medicine distributors. Full CRUD with referential integrity on deletion — if medicines are linked to a supplier, the delete is rejected with a clear error code to protect inventory data.

GET	**http://localhost:8080/api/suppliers**	*List all suppliers with medicine count*

```
{
  "status": "success",
  "count":  3,
  "data": [
    { "supplierId": 1, "supplierName": "Apollo Distributors", "contactPerson":
"Rajesh Gupta", "totalMedicines": 2 },
    { "supplierId": 2, "supplierName": "MedPlus Supply Chain", "contactPerson":
"Suresh Nair",  "totalMedicines": 2 },
    { "supplierId": 3, "supplierName": "Himalaya Wellness",    "contactPerson":
"Anjali Mehta", "totalMedicines": 1 }
  ]
}
```

GET	**http://localhost:8080/api/suppliers/{supplierId}**	*Supplier profile with all medicines and average price*

```
{
  "status": "success",
  "data": {
    "supplierId":         1,
    "supplierName":       "Apollo Distributors",
    "contactPerson":      "Rajesh Gupta",
    "phone":              "022-123456",
    "averageMedicinePrice": 75.00,
    "medicines": [
      { "medicineId": 1, "name": "Dolo 650",    "category": "Tablet", "price": 30.00
},
      { "medicineId": 2, "name": "Azithral 500","category": "Tablet", "price": 120.00
}
    ]
  }
}
```

POST	http://localhost:8080/api/suppliers	*Register a new supplier*

Request Body

```
{
  "supplierName":   "Sun Pharma Distributors",
  "contactPerson": "Meena Joshi",
  "phone":          "080-556677"
}
```

201 Created	*Response Body*

```
{
  "status":  "success",
  "message": "Supplier registered successfully",
  "data":    { "supplierId": 4, "supplierName": "Sun Pharma Distributors" }
}
```

PUT	http://localhost:8080/api/suppliers/{supplierId}	*Update supplier name or contact details*

Request Body

```
{
  "supplierName":   "Sun Pharma Distributors Pvt Ltd",
  "contactPerson": "Meena Joshi",
  "phone":          "080-998877"
}
```

200 OK	*Response Body*

```
{ "status": "success", "message": "Supplier updated", "data": { "supplierId": 4 } }
```

DELETE	**http://localhost:8080/api/suppliers/{supplierId}**	*Remove supplier — blocked if medicines are linked*

200 OK	*Response Body*

```
{ "status": "success", "message": "Supplier with ID 4 deleted successfully" }
```

409 Conflict	*Response Body*

```
{
  "status":  "error",
  "code":    "SUPPLIER_HAS_MEDICINES",
  "message": "Cannot delete supplier with existing medicines. Reassign medicines first."
}
```

GET	**http://localhost:8080/api/suppliers/{supplierId}/medicines**	*All medicines supplied by this supplier*

Sample: GET /api/suppliers/1/medicines

```
{
  "status":       "success",
  "supplierId":    1,
  "supplierName": "Apollo Distributors",
  "count":         2,
  "data": [
    { "medicineId": 1, "name": "Dolo 650",     "category": "Tablet", "price": 30.00,
"stockQuantity": 500 },
    { "medicineId": 2, "name": "Azithral 500", "category": "Tablet", "price": 120.00,
"stockQuantity": 50  }
  ]
}
```

Domain 3 — Medicines

`Base path:  http://localhost:8080/api/medicines`

The Medicines domain is the heart of the inventory. It covers the full lifecycle of a medicine entry — creation, retrieval, update, removal — as well as business-specific queries like expiry alerts, category filters, and price-range searches.

GET	**http://localhost:8080/api/medicines**	*List all medicines — filterable and paginated*

Query params: page, size, sort, order, category, supplierId, minPrice, maxPrice

Sample: GET /api/medicines?category=Tablet&page=0&size=10

```
{
  "status":        "success",
  "page":          0,
  "size":          10,
  "totalElements": 3,
  "totalPages":    1,
  "data": [
    {
      "medicineId":    1,
      "name":          "Dolo 650",
      "category":      "Tablet",
      "price":         30.00,
      "stockQuantity": 500,
      "expiryDate":    "2025-12-31",
      "supplier":      { "supplierId": 1, "supplierName": "Apollo Distributors" },
      "inStock":       true
    }
  ]
}
```


GET	**http://localhost:8080/api/medicines/{medicineId}**	*Get full details of a single medicine*

200 OK	*Response Body*

```
{
  "status": "success",
  "data": {
    "medicineId":    1,
    "name":          "Dolo 650",
    "category":      "Tablet",
    "price":         30.00,
    "stockQuantity": 500,
    "expiryDate":    "2025-12-31",
    "inStock":       true,
    "supplier": {
      "supplierId":    1,
      "supplierName":  "Apollo Distributors",
      "contactPerson": "Rajesh Gupta",
      "phone":         "022-123456"
    }
  }
}
```

404 Not Found	*Response Body*

```
{
  "status":  "error",
  "code":    "MEDICINE_NOT_FOUND",
  "message": "No medicine found with ID: 99"
}
```

POST	http://localhost:8080/api/medicines	*Add a new medicine to the inventory*

Request Body

```
{
  "name":          "Calpol 500",
  "category":      "Tablet",
  "price":         35.00,
  "stockQuantity": 300,
  "expiryDate":    "2026-06-30",
  "supplierId":    3
}
```

201 Created	*Response Body*

```
{
  "status":  "success",
  "message": "Medicine added to inventory successfully",
  "data": {
    "medicineId":    6,
    "name":          "Calpol 500",
    "category":      "Tablet",
    "price":         35.00,
    "stockQuantity": 300,
    "expiryDate":    "2026-06-30",
    "supplierId":    3
  }
}
```

PUT	http://localhost:8080/api/medicines/{medicineId}	*Update medicine details — price, stock, expiry*

Request Body

```
{
  "name":          "Calpol 500mg",
  "category":      "Tablet",
  "price":         38.00,
  "stockQuantity": 300,
  "expiryDate":    "2026-06-30",
  "supplierId":    3
}
```

200 OK	*Response Body*

```
{
  "status":  "success",
  "message": "Medicine updated successfully",
  "data":    { "medicineId": 6, "price": 38.00, "stockQuantity": 300 }
}
```

DELETE	http://localhost:8080/api/medicines/{medicineId}	*Remove a medicine from inventory*

200 OK	*Response Body*

```
{ "status": "success", "message": "Medicine with ID 6 removed from inventory" }
```

GET	http://localhost:8080/api/medicines/by-category/{category}	*All medicines in a specific category — category pages*

Sample: GET /api/medicines/by-category/Tablet

```
{
  "status":   "success",
  "category": "Tablet",
  "count":    3,
  "data": [
    { "medicineId": 1, "name": "Dolo 650",     "price": 30.00,  "supplier": "Apollo Distributors" },
    { "medicineId": 2, "name": "Azithral 500", "price": 120.00, "supplier": "Apollo Distributors" },
    { "medicineId": 4, "name": "Combiflam",    "price": 45.00,  "supplier": "Himalaya Wellness"   }
  ]
}
```

GET	http://localhost:8080/api/medicines/by-supplier/{supplierId}	*All medicines from a specific supplier — supplier pages*

Sample: GET /api/medicines/by-supplier/2

```
{
  "status":       "success",
  "supplierId":    2,
  "supplierName": "MedPlus Supply Chain",
  "count":         2,
  "data": [
    { "medicineId": 3, "name": "Benadryl",   "category": "Syrup",     "price": 110.00 },
    { "medicineId": 5, "name": "Insulin Pen", "category": "Injection", "price": 800.00 }
  ]
}
```

GET	**http://localhost:8080/api/medicines/price-range**	*Filter medicines between min and max price*

Query params: min, max

Sample: GET /api/medicines/price-range?min=30&max=150

```
{
  "status":   "success",
  "minPrice": 30.00,
  "maxPrice": 150.00,
  "count":    4,
  "data": [
    { "medicineId": 1, "name": "Dolo 650",    "price": 30.00  },
    { "medicineId": 3, "name": "Benadryl",    "price": 110.00 },
    { "medicineId": 4, "name": "Combiflam",   "price": 45.00  },
    { "medicineId": 2, "name": "Azithral 500","price": 120.00 }
  ]
}
```

Domain 4 — Orders

`Base path:  http://localhost:8080/api/orders`

The most business-critical domain. Order placement must validate stock and deduct quantities in a single atomic transaction. Cancellation restores stock. The invoice endpoint returns a payment-ready JSON payload for print or digital billing.

POST	http://localhost:8080/api/orders	*Place a new order — validates stock, deducts quantities*

Request Body

```
{
  "customerId":  1,
  "paymentMode": "UPI",
  "items": [
    { "medicineId": 1, "quantity": 5 },
    { "medicineId": 4, "quantity": 2 }
  ]
}
```

201 Created	*Response Body*

```
{
  "status":  "success",
  "message": "Order placed successfully",
  "data": {
    "orderId":     6,
    "orderDate":   "2024-06-10",
    "customerId":  1,
    "paymentMode": "UPI",
    "orderTotal":  240.00,
    "items": [
      {
        "medicineId":   1,
        "medicineName": "Dolo 650",
        "quantity":     5,
        "priceAtOrder": 30.00,
        "subtotal":     150.00
      },
      {
        "medicineId":   4,
        "medicineName": "Combiflam",
        "quantity":     2,
        "priceAtOrder": 45.00,
        "subtotal":     90.00
      }
    ]
  }
}
```


409 Conflict — insufficient stock	*Response Body*

```
{
  "status":  "error",
  "code":    "INSUFFICIENT_STOCK",
  "message": "Requested 10 units of Insulin Pen but only 5 in stock."
}
```

GET	**http://localhost:8080/api/orders/{orderId}**	*Full order detail with line items — order confirmation*

```
{
  "status": "success",
  "data": {
    "orderId":     1,
    "orderDate":   "2023-10-01",
    "paymentMode": "UPI",
    "customer": {
      "customerId": 1, "name": "Rahul Sharma",
      "phone": "9876543210", "city": "Mumbai"
    },
    "items": [
      {
        "itemId":       1,
        "medicine":     { "medicineId": 1, "name": "Dolo 650", "category":
"Tablet" },
        "quantity":     5,
        "priceAtOrder": 30.00,
        "subtotal":     150.00
      }
    ],
    "orderTotal": 150.00
  }
}
```

GET	**http://localhost:8080/api/orders**	*Admin: all orders with date-range filter and pagination*

Query params: from, to (date), page, size, paymentMode

Sample: GET /api/orders?from=2023-10-01&to=2023-10-31

```
{
  "status":        "success",
  "totalElements": 5,
  "data": [
    { "orderId": 1, "orderDate": "2023-10-01", "customerName": "Rahul Sharma",
"orderTotal": 150.00,  "paymentMode": "UPI"  },
    { "orderId": 2, "orderDate": "2023-10-02", "customerName": "Priya Verma",
"orderTotal": 240.00,  "paymentMode": "Card" },
    { "orderId": 5, "orderDate": "2023-10-12", "customerName": "Sneha Reddy",
"orderTotal": 1190.00, "paymentMode": "Card" }
  ]
}
```

GET	**http://localhost:8080/api/orders/by-date-range**	*Orders in a calendar window — monthly sales reports*

Query params: from, to

```
{
  "status": "success",
  "period": { "from": "2023-10-01", "to": "2023-10-31" },
  "totalOrders":  5,
  "totalRevenue": 2435.00,
  "data": [
    { "orderId": 1, "orderDate": "2023-10-01", "customerName": "Rahul Sharma",
"orderTotal": 150.00 },
    { "orderId": 5, "orderDate": "2023-10-12", "customerName": "Sneha Reddy",
"orderTotal": 1190.00 }
  ]
}
```

DELETE	**http://localhost:8080/api/orders/{orderId}**	*Cancel order — restores stock quantities automatically*

200 OK	*Response Body*

```
{ "status": "success", "message": "Order 1 cancelled. Stock quantities restored." }
```

GET	**http://localhost:8080/api/orders/by-payment-mode/{mode}**	*All orders for a given payment mode*

Sample: GET /api/orders/by-payment-mode/UPI

```
{
  "status":       "success",
  "paymentMode":  "UPI",
  "totalOrders":  2,
  "totalRevenue": 950.00,
  "data": [
    { "orderId": 1, "orderDate": "2023-10-01", "customerName": "Rahul Sharma",
"orderTotal": 150.00 },
    { "orderId": 4, "orderDate": "2023-10-10", "customerName": "Rahul Sharma",
"orderTotal": 800.00 }
  ]
}
```

GET	**http://localhost:8080/api/orders/{orderId}/invoice**	*Invoice-ready JSON for a completed order*

```
{
  "status": "success",
  "invoice": {
    "invoiceNumber": "JRP-INV-00001",
    "orderId":        1,
    "orderDate":      "2023-10-01",
    "paymentMode":    "UPI",
    "billedTo": { "name": "Rahul Sharma", "phone": "9876543210", "city": "Mumbai" },
    "items": [
      { "medicineName": "Dolo 650", "qty": 5, "unitPrice": 30.00, "total": 150.00 }
    ],
    "subTotal":    150.00,
    "gst":           0.00,
    "grandTotal":  150.00,
    "generatedAt": "2024-06-10T12:30:00"
  }
}
```

Domain 5 — Inventory

`Base path:  http://localhost:8080/api/inventory`

Pharmacy staff and admin users get a real-time view of stock levels. Low-stock and expiry alerts feed reorder workflows. The PATCH restock endpoint adds a quantity delta and captures a reason for audit trails. The PATCH endpoint is used (not PUT) since only stock quantity is modified.

GET	**http://localhost:8080/api/inventory**	*Complete stock overview for all medicines*


```
{
  "status": "success",
  "data": [
    { "medicineId": 1, "name": "Dolo 650",     "stockQuantity": 500, "expiryDate": "2025-12-31", "stockStatus": "IN_STOCK"  },
    { "medicineId": 2, "name": "Azithral 500", "stockQuantity": 50,  "expiryDate": "2024-05-20", "stockStatus": "LOW_STOCK" },
    { "medicineId": 3, "name": "Benadryl",     "stockQuantity": 20,  "expiryDate": "2024-11-15", "stockStatus": "LOW_STOCK" },
    { "medicineId": 4, "name": "Combiflam",    "stockQuantity": 200, "expiryDate": "2026-01-01", "stockStatus": "IN_STOCK"  },
    { "medicineId": 5, "name": "Insulin Pen",  "stockQuantity": 5,   "expiryDate": "2024-03-10", "stockStatus": "CRITICAL"  }
  ]
}
```


GET	**http://localhost:8080/api/inventory/low-stock**	*Medicines below a stock threshold — triggers restock alerts*

Query param: threshold (default 50)

Sample: GET /api/inventory/low-stock?threshold=50

```
{
  "status":     "success",
  "threshold":  50,
  "alertCount": 3,
  "data": [
    {
      "medicineId":    2,
      "name":          "Azithral 500",
      "category":      "Tablet",
      "stockQuantity": 50,
      "supplierName":  "Apollo Distributors",
      "supplierPhone": "022-123456",
      "message":       "Stock below threshold. Reorder recommended."
    },
    {
      "medicineId":    3,
      "name":          "Benadryl",
      "category":      "Syrup",
      "stockQuantity": 20,
      "supplierName":  "MedPlus Supply Chain",
      "supplierPhone": "040-987654",
      "message":       "Stock below threshold. Reorder recommended."
    }
  ]
}
```

PATCH	**http://localhost:8080/api/inventory/{medicineId}/restock**	*Update stock after a new shipment — delta quantity*

Request Body

```
{
  "additionalQuantity": 200,
  "reason":             "New shipment received from Apollo Distributors"
}
```

200 OK	*Response Body*

```
{
  "status":  "success",
  "message": "Stock updated for Azithral 500",
  "data": {
    "medicineId":         2,
    "previousStock":      50,
    "additionalQuantity": 200,
    "newStock":           250,
    "updatedAt":          "2024-06-10T14:00:00"
  }
}
```

Domain 6 — Analytics & Reports

`Base path:  http://localhost:8080/api/reports`

All endpoints are read-only. They aggregate data across all tables to power management dashboards, supplier performance analysis, loyalty programs, and compliance audits. Use date-range parameters to generate period-specific reports.

GET	**http://localhost:8080/api/reports/revenue**	*Total revenue over a date range — core business KPI*

Query params: from, to

Sample: GET /api/reports/revenue?from=2023-01-01&to=2023-12-31

```
{
  "status": "success",
  "period": { "from": "2023-01-01", "to": "2023-12-31" },
  "totalOrders":       5,
  "totalRevenue":      2435.00,
  "averageOrderValue": 487.00
}
```

GET	**http://localhost:8080/api/reports/revenue-by-payment-mode**	*Revenue breakdown per payment mode — cash-flow analysis*

```
{
  "status": "success",
  "data": [
    { "paymentMode": "Card", "totalOrders": 2, "totalRevenue": 1440.00,
"avgOrderValue": 720.00 },
    { "paymentMode": "UPI",  "totalOrders": 2, "totalRevenue":  950.00,
"avgOrderValue": 475.00 },
    { "paymentMode": "Cash", "totalOrders": 1, "totalRevenue":   45.00,
"avgOrderValue":  45.00 }
  ],
  "grandTotal": 2435.00
}
```


GET	**http://localhost:8080/api/reports/bestsellers**	*Top medicines by quantity sold — full analytics data*

Query params: limit, from, to


```
{
  "status": "success",
  "data": [
    { "medicineId": 1, "name": "Dolo 650",    "category": "Tablet",
"totalQuantitySold": 10, "totalRevenue": 300.00  },
    { "medicineId": 2, "name": "Azithral 500","category": "Tablet",
"totalQuantitySold":  4, "totalRevenue": 480.00  },
    { "medicineId": 5, "name": "Insulin Pen", "category": "Injection",
"totalQuantitySold":  2, "totalRevenue": 1600.00 }
  ]
}
```


GET	**http://localhost:8080/api/reports/customer-with-most-orders**	*Customer with most orders — VIP identification*

```
{
  "status": "success",
  "data": {
    "customerId":       1,
    "name":             "Rahul Sharma",
    "phone":            "9876543210",
    "city":             "Mumbai",
    "totalOrders":      2,
    "totalAmountSpent": 950.00
  }
}
```

GET	**http://localhost:8080/api/reports/expired-medicines**	*Medicines past expiry date — compliance audit*

```
{
  "status":       "success",
  "auditDate":    "2024-06-10",
  "totalExpired": 1,
  "data": [
    {
      "medicineId":     5,
      "name":           "Insulin Pen",
      "category":       "Injection",
      "expiryDate":     "2024-03-10",
      "daysSinceExpiry": 92,
      "stockQuantity":  5,
      "estimatedLoss":  4000.00,
      "supplierName":   "MedPlus Supply Chain"
    }
  ]
}
```

GET	**http://localhost:8080/api/reports/inventory-audit**	*Low-stock medicines with supplier contact — reorder sheet*

Query param: threshold (default 50)

```
{
  "status":           "success",
  "threshold":        50,
  "totalLowStockItems": 3,
  "data": [
    {
      "medicineId":    2,
      "medicineName":  "Azithral 500",
      "stockQuantity": 50,
      "expiryDate":    "2024-05-20",
      "supplierName":  "Apollo Distributors",
      "supplierPhone": "022-123456",
      "contactPerson": "Rajesh Gupta"
    },
    {
      "medicineId":    5,
      "medicineName":  "Insulin Pen",
      "stockQuantity": 5,
      "expiryDate":    "2024-03-10",
      "supplierName":  "MedPlus Supply Chain",
      "supplierPhone": "040-987654",
      "contactPerson": "Suresh Nair"
    }
  ]
}
```


Domain 7 — Global Search

`Base path: http://localhost:8080/api/search`

A single cross-entity endpoint powering the pharmacy search bar. Queries both Medicines (by name) and Suppliers (by name) in one call. The type parameter narrows scope to medicines, suppliers, or all entities.

GET	**http://localhost:8080/api/search**	*Cross-entity full-text search across medicines and suppliers*

Query params: q (keyword), type (all | medicines | suppliers)

Sample: GET /api/search?q=dolo&type=all


```
{
  "status": "success",
  "query":   "dolo",
  "results": {
    "medicines": [
      { "medicineId": 1, "name": "Dolo 650", "category": "Tablet", "price": 30.00,
"supplier": "Apollo Distributors" }
    ],
    "suppliers": [],
    "totalHits": 1
  }
}
```


Complete Endpoint Summary

All 38 endpoints consolidated in a single reference table.

Method	Endpoint URL	Purpose
GET	/api/customers	List all customers — filterable and paginated
GET	/api/customers/{customerId}	Get customer profile with order summary
POST	/api/customers	Register a new customer
PUT	/api/customers/{customerId}	Update customer profile
DELETE	/api/customers/{customerId}	Remove customer (integrity check)
GET	/api/customers/{customerId}/orders	Full order history for a customer
GET	/api/customers/top-spenders	Top customers by total amount spent
GET	/api/suppliers	List all suppliers with medicine count
GET	/api/suppliers/{supplierId}	Supplier profile with medicine list
POST	/api/suppliers	Register a new supplier
PUT	/api/suppliers/{supplierId}	Update supplier information
DELETE	/api/suppliers/{supplierId}	Remove supplier (integrity check)

Method	Endpoint URL	Purpose
GET	/api/suppliers/{supplierId}/medicines	Medicines supplied by this supplier
GET	/api/medicines	List all medicines — filterable and paginated
GET	/api/medicines/{medicineId}	Get a single medicine with full details
POST	/api/medicines	Add a new medicine to inventory
PUT	/api/medicines/{medicineId}	Update medicine details
DELETE	/api/medicines/{medicineId}	Remove a medicine from inventory
GET	/api/medicines/by-category/{category}	All medicines in a specific category
GET	/api/medicines/by-supplier/{supplierId}	All medicines by a specific supplier
GET	/api/medicines/price-range	Filter medicines between min and max price
POST	/api/orders	Place new order (stock check + deduction)
GET	/api/orders/{orderId}	Full order detail with line items
GET	/api/orders	Admin: all orders with date filter
GET	/api/orders/by-date-range	Orders within a calendar window
DELETE	/api/orders/{orderId}	Cancel order — restores stock
GET	/api/orders/by-payment-mode/{mode}	Orders filtered by payment mode
GET	/api/orders/{orderId}/invoice	Invoice-ready JSON for an order
GET	/api/inventory	Complete stock overview
GET	/api/inventory/low-stock	Medicines below stock threshold
PATCH	/api/inventory/{medicineId}/restock	Restock a medicine by delta quantity
GET	/api/reports/revenue	Total revenue over a date range
GET	/api/reports/revenue-by-payment-mode	Revenue breakdown by payment mode
GET	/api/reports/bestsellers	Top medicines by quantity sold
GET	/api/reports/customer-with-most-orders	Customer with highest order count
GET	/api/reports/expired-medicines	Medicines past their expiry date
GET	/api/reports/inventory-audit	Low-stock medicines with supplier contact
GET	/api/search	Cross-entity search — medicines and suppliers

Architectural Notes

- Use PATCH (not PUT) on /inventory restock — only stockQuantity changes; PUT implies full resource replacement.

- POST /api/orders must validate stock availability and deduct quantities in a single DB transaction to prevent race conditions under concurrent load.
- The priceAtOrder column in order_items captures the price at the time of sale — order history never joins against the live medicines.price to avoid retroactive pricing discrepancies.
- All list endpoints must be paginated from day one using the standard page, size, sort, order query parameters.
- Referential integrity violations (SUPPLIER_HAS_MEDICINES, MEDICINE_IN_ORDER, CUSTOMER_HAS_ORDERS) must be caught at the service layer — never surfaced as raw database exceptions.
- DELETE /api/orders must restore stock within the same transaction to keep inventory consistent across all medicine records.
- Expired medicines (/api/reports/expired-medicines) must be quarantined immediately; the endpoint is the audit trigger for CDSCO compliance workflows.